



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PRAKASH AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

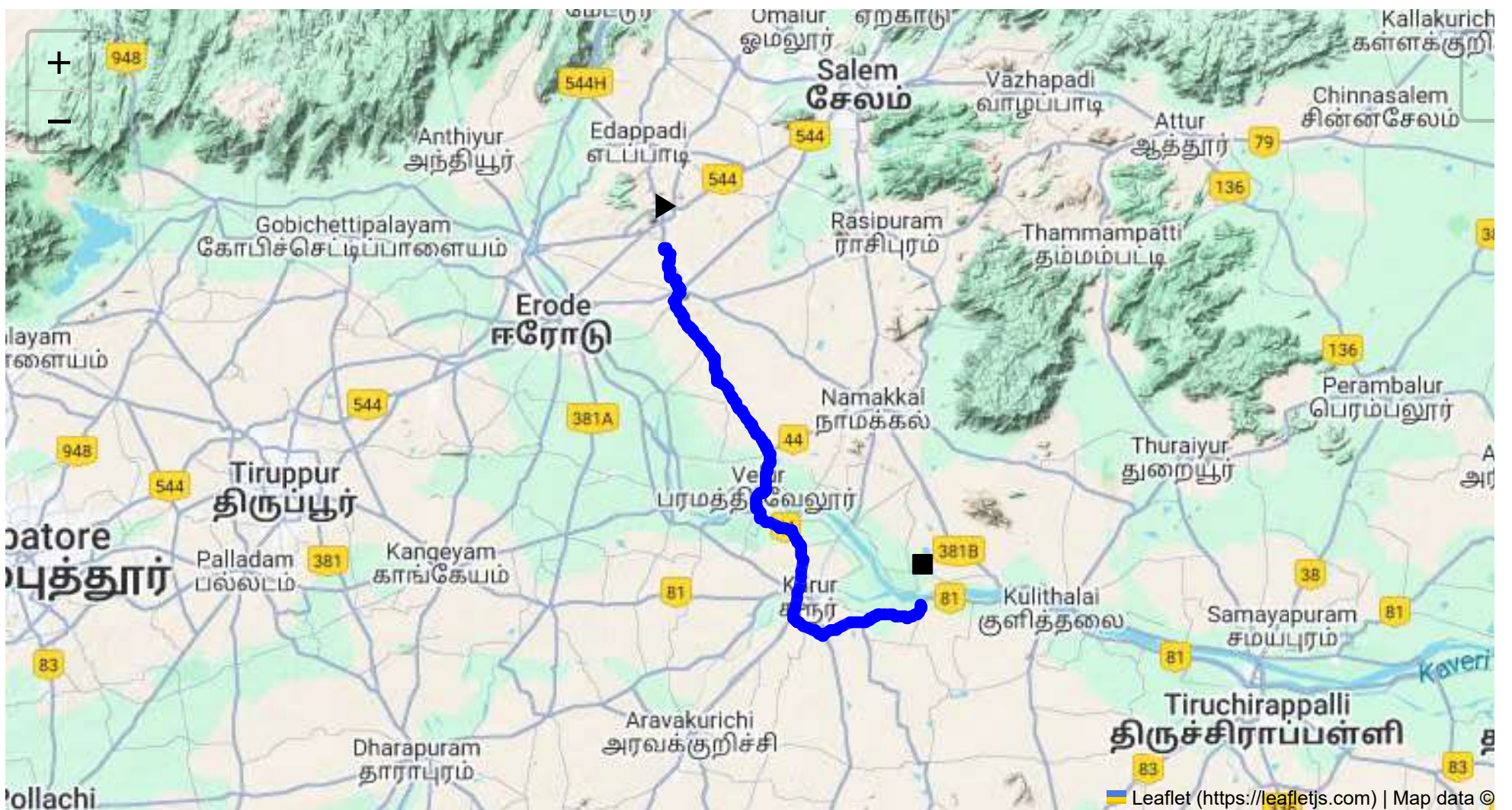
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 90.97 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (10.943356, 78.232684)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to W6VM+83C, Karur, Tamil Nadu spans approximately 91 kilometers. It usually follows National Highway 544 (NH 544) and National Highway 81 (NH 81), with a mix of rural and semi-urban landscapes. The primary course is generally straightforward but requires navigation through some intersecting roads in semi-urban areas.

Typical Weather Conditions

Tamil Nadu experiences a tropical climate:

- **Summer (March to June):** High temperatures, can exceed 40°C. Heat can affect vehicle performance.
- **Monsoon (July to September):** Heavy rainfall could lead to reduced visibility and slippery roads.
- **Winter (October to February):** Milder temperatures, but occasional fog could occur, especially during early morning hours.

Traffic Patterns

- **Peak Hours:** Typically between 8-10 AM and 5-7 PM.
- **Congestion-Prone Areas:** Expect higher traffic density near major towns such as Tiruchengode and Namakkal.
- **Holidays/Festivals:** Traffic can be particularly heavy around major holidays like Pongal and Diwali.

Road Quality and Infrastructure

- **NH 544 and NH 81:** Generally good condition, but occasional potholes and uneven surfaces might be present due to frequent wear and tear.
- **Local Roads:** Some connecting and local roads might be narrower and less maintained.

Alternative Routes for Emergencies

- **State Highways 79 and 95:** Provide alternative paths though longer and possibly less direct.
- **Local Routes:** Function as detours but condition and suitability for heavy vehicles need to be verified.

Local Regulations on Hazardous Material Transport

- **Permits Required:** Special permits needed for transporting hazardous materials.
- **Restricted Timings:** Some towns may impose restrictions on entry times for trucks carrying hazardous materials to minimize traffic disruptions.

Historical Incidents

- **Accidents Reports:** Occasional reports of truck rollovers or collisions due to narrow roads and sharp turns, particularly near congested towns.
- **Hazard Material Spills:** Rare but have occurred, emphasizing need for adherence to safety regulations.

Environmental Considerations

- **Sensitive Areas:** Proximity to water bodies like the Kaveri River requires careful handling to prevent contamination.
- **Wildlife:** Areas near forested zones may have wildlife crossings.

Communication Coverage

- **Dead Zones:** Some rural stretches might have patchy mobile network coverage, particularly when moving through hilly terrains or remote areas.
- **Recommendations:** Equip trucks with GPS and communication devices that can operate on multiple network technologies.

Estimated Emergency Response Times

- **Urban Areas:** 15-30 minutes, thanks to proximity to hospitals and fire stations.
- **Rural Sections:** Could exceed 30 minutes, especially during inclement weather.

Overall Risk Assessment Summary

- **Primary Risks:** Weather (especially monsoon), traffic congestion, road quality variances, and communication dead zones.
- **Safety Measures:** Ensure vehicle roadworthiness, proper documentation, adherence to regulations, and availability of emergency communication tools.
- **Emergency Planning:** Pre-plan alternative routes and maintain a list of emergency contacts.

The overall risk level is moderate, with appropriate caution and preparedness required to ensure safety and compliance on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	10.94726, 78.23412	15 KM/Hr
13	Turn	Medium	10.94755, 78.23388	30 KM/Hr
14	Blind Spot	Blind Spot	10.94814, 78.23384	10 KM/Hr
15	Turn	High	10.94786, 78.23439	15 KM/Hr
16	Turn	High	10.94362, 78.23248	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
19	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
20	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94726, 78.23412



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.94755, 78.23388



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.94786, 78.23439



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.94362, 78.23248
